

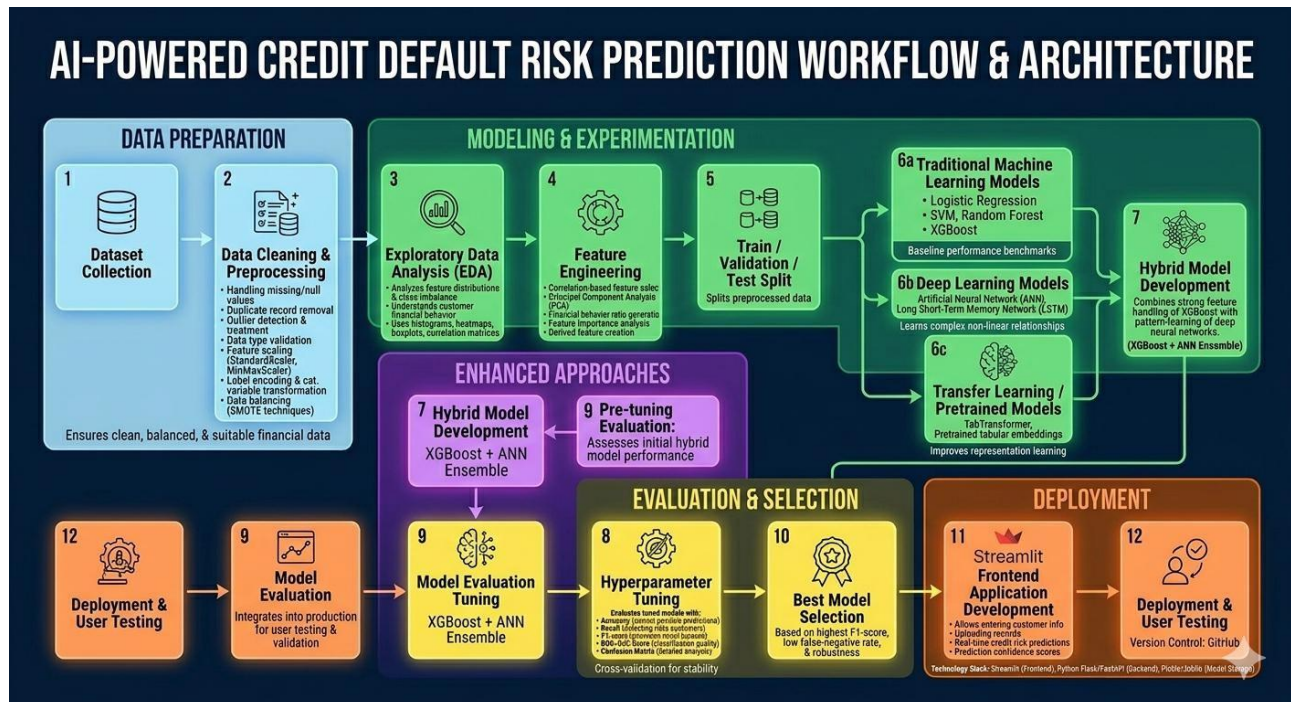
Project Topic – AI Powered Credit Default Risk Prediction System using Machine Learning and Neural Network Techniques

Pipeline Overview:

The project pipeline is designed to develop an AI-powered credit default risk prediction system using Machine Learning and Deep Learning techniques. The workflow includes data collection, preprocessing, feature engineering, model training, hyperparameter optimisation, model evaluation, and deployment of the final predictive application.

The proposed workflow ensures efficient handling of financial datasets, improves prediction accuracy, and supports real-world application deployment for financial risk assessment.

Workflow Diagram:



Data Pre-processing and Feature Engineering:

Data preprocessing is a critical stage of the project because it prepares the datasets for effective Machine Learning and Deep Learning model development. During this stage, the financial datasets will be cleaned, transformed, and standardized to improve data quality and prediction performance. These processes help ensure that the datasets are reliable, consistent, and suitable for accurate credit default risk prediction. The preprocessing steps include:

- Handling missing and null values
- Removing duplicate records
- Detecting and treating outliers
- Validating and correcting data types
- Feature scaling using StandardScaler and MinMaxScaler
- Encoding categorical variables using Label Encoding and One-Hot Encoding
- Evaluating and comparing class balancing techniques, including No Balancing, SMOTE, ADASYN, and Random Oversampling, to determine their impact on model performance
- Splitting the datasets into training, validation, and testing sets

These preprocessing techniques ensure that the datasets are clean, balanced, standardised, and suitable for accurate credit risk prediction and financial behaviour analysis.

Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) will be conducted to understand the structure and behaviour of the financial datasets before model development. The analysis will focus on identifying feature distributions, customer financial behaviour patterns, class imbalance, correlations between variables, and credit default trends.

The following visualisation techniques will be used:

- Histograms
- Heatmaps
- Boxplots
- Correlation matrices
- Distribution plots

Each visualization technique will support specific analytical objectives. Histograms and distribution plots will be used to understand feature distributions and identify skewed variables. Boxplots will assist in detecting outliers, while heatmaps and correlation matrices will reveal relationships between financial variables and the target class. These insights will guide feature engineering decisions and help identify factors associated with credit default risk.

Feature Engineering:

Feature engineering techniques will be applied to improve the predictive performance of the machine learning and deep learning models. The process will focus on transforming raw financial data into meaningful features that better represent customer credit behaviour.

The feature engineering process includes :

- Correlation-based feature selection
- Principal Component Analysis (PCA)
- Financial behaviour ratio generation
- Feature importance analysis
- Feature normalisation and scaling
- Derived feature creation

Feature engineering techniques will be evaluated based on their contribution to predictive performance. Correlation analysis will help remove redundant variables, while PCA will be investigated as a dimensionality reduction technique to improve computational efficiency. Derived financial indicators and behavioural ratios will be created to capture customer credit patterns that may not be directly observable in the original datasets.

Model Training and Improvement:

The project will develop and evaluate multiple Machine Learning and Deep Learning approaches to determine the most effective model for predicting customer credit default risk. Traditional classification algorithms such as Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost will first be implemented as benchmark models for structured financial datasets. These baseline models will provide a performance reference for comparing more advanced AI techniques.

To support responsible AI development, the project will incorporate bias detection and fairness evaluation throughout the model development process. Since financial datasets may contain attributes that indirectly influence lending decisions, fairness assessment will be performed to identify and minimise potential discriminatory outcomes. The project will evaluate fairness using metrics such as Demographic Parity, Equal Opportunity, Equalized Odds, and Disparate Impact Ratio. Fairness evaluation will be implemented using Fairlearn and IBM AIF360 frameworks.

To better capture complex customer financial behaviour and hidden non-linear relationships within the data, Deep Learning architectures including Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks will also be explored. Furthermore, the project will investigate transfer learning methods, including TabTransformer and pretrained tabular embedding models, to enhance feature representation and improve prediction capability for structured financial data.

A hybrid ensemble model combining XGBoost and Artificial Neural Networks (ANN) will be proposed to improve prediction accuracy and model robustness. The hybrid model aims to combine the powerful feature-handling capability of XGBoost with the advanced pattern-learning ability of deep neural networks.

To further optimise model performance, several hyperparameter tuning and optimisation techniques will be applied, including GridSearchCV, Random Search, Stratified K-Fold Cross-Validation (k=5), learning rate tuning, batch size optimisation, epoch tuning, dropout regularisation, and early stopping. These optimisation techniques will help improve prediction accuracy, reduce overfitting, and enhance the generalisation capability of the model.

Model Evaluation and Metrics:

The performance of the trained models will be assessed using several classification evaluation metrics to measure prediction quality, reliability, and effectiveness. Metrics such as **Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix analysis** will be used to evaluate how well each model identifies potential credit default customers. Accuracy will measure overall prediction performance, while Precision and Recall will assess the model's ability to correctly classify risky customers. The F1-Score will provide a balanced measure of Precision and Recall, and the ROC-AUC Score will evaluate the model's overall classification strength. Confusion Matrix analysis will also be used to examine correct and incorrect predictions in detail.

In addition, cross-validation techniques will be applied to test model consistency and generalisation across different data samples. The final model selection will be based on overall robustness, strong F1-Score performance, balanced Precision and Recall values, and a low false-negative rate to minimise the risk of missing high-risk customers.

Application Deployment:

The final AI model will be integrated into a user-friendly web application developed using Streamlit. The application will allow users to enter customer financial information, upload customer financial records, receive real-time credit risk predictions, and view prediction confidence scores generated by the trained AI model. The deployed application will also include explainability capabilities that allow users to understand the key factors contributing to each prediction. This functionality improves transparency, supports regulatory compliance, and promotes responsible use of AI within financial decision-making environments. The deployment architecture consists of multiple components that work together to deliver a complete AI-powered prediction system.

- **Frontend – Streamlit:**
Streamlit will be used to develop the graphical user interface (GUI) of the application. It provides an interactive web-based platform where users can input customer financial details and view prediction results in real time. Streamlit is lightweight, easy to deploy, and suitable for rapid AI application development.
- **Backend – Python Flask / FastAPI:**
Flask or FastAPI will be used to handle backend processing and communication between the frontend and the trained machine learning models. The backend will process user inputs, send data to the prediction model, and return the prediction results to the application interface.
- **Model Storage – Pickle / Joblib:**
The trained machine learning and deep learning models will be stored using Pickle or Joblib serialization techniques. This allows the saved models to be loaded efficiently during deployment without retraining the models each time the application runs.
- **Version Control – GitHub:**
GitHub will be used for project version control, team collaboration, code management, and documentation. It enables all team members to manage updates, track changes, review code, and maintain organised project development throughout the project lifecycle.

Final Workflow Summary:

The proposed pipeline workflow combines data preprocessing, feature engineering, machine learning, deep learning, transfer learning, hybrid modelling, optimisation, and deployment into a complete AI-driven credit risk prediction framework. The workflow is designed to ensure, high model performance, scalability, ethical AI usage, fairness evaluation, bias detection, explainable decision-making and practical real-world application.